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Full length article

AttenScribble: Attention-enhanced scribble supervision for medical image segmentation[☆]

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ABSTRACT

The success of deep networks in medical image segmentation relies heavily on massive labeled training data. However, acquiring dense annotations is a time-consuming process. Weakly supervised methods normally employ less expensive forms of supervision, among which scribbles started to gain popularity lately thanks to their flexibility. However, due to the lack of shape and boundary information, it is extremely challenging to train a deep network on scribbles that generalize on unlabeled pixels. In this paper, we present a straightforward yet effective scribble-supervised learning framework. Inspired by recent advances in transformer-based segmentation, we create a pluggable spatial self-attention module that could be attached on top of any internal feature layers of arbitrary fully convolutional network (FCN) backbone. The module infuses global interaction while keeping the efficiency of convolutions. Descended from this module, we construct a similarity metric based on normalized and symmetrized attention. This attentive similarity leads to a novel regularization loss that imposes consistency between segmentation prediction and visual affinity. This attentive similarity loss optimizes the alignment of FCN encoders, attention mapping and model prediction. Ultimately, the proposed FCN+Attention architecture can be trained end-to-end guided by a combination of three learning objectives: partial segmentation loss, customized masked conditional random fields, and the proposed attentive similarity loss. Extensive experiments on public datasets (ACDC and CHAOS) showed that our framework not only outperforms existing state-of-the-art but also delivers close performance to fully-supervised benchmarks. The code is available at https://github.com/YangQinzhu/AttenScribble.git.

1. Introduction

Deep networks have demonstrated remarkable advancements in medical image segmentation [1]. Nonetheless, their capacity to generalize well on unseen examples relies heavily on the quantity of manually annotated training data. Obtaining sufficient labeled data can be challenging, especially for medical image segmentation, where acquiring dense annotations requires extensive expertise.

To mitigate this burden, weakly-supervised methods have been proposed that employ less expensive forms of supervision [2–5], such as scribbles [6–8], boxes [9], points [10], or image-level labels [11]. Scribble supervision has become increasingly popular lately since it allows more flexible and natural annotations while providing reasonable initial target locations. However, due to insufficient shape and boundary information, it is extremely challenging to train a deep network on scribbles that accurately generates predictions on unlabeled pixels.

One common remedy would be to leverage pseudo labels [12], that can be generated via graph-based segmentation such as random walker [13] to propagate scribble annotations to unlabeled pixels. Lately, more approaches tried to exploit model output to produce reliable pseudo supervision signals. For example, Scribble2Label [14] proposed to consolidate model predictions along different training iterations; CycleMix [15] adopted a mixed-up strategy with consistency regularization to improve data quality. A more recent work [6] introduced an elegantly designed dual-branch network with mixed-up strategies that achieved new state-of-the-art. Fusing predictions from separate branches introduced diversified information and thus

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greatly increased robustness. In contrast, ScribbleVC [16] and ScribFormer [17] leverage CNN- and Transformer-based features to further enhance model performance. Aside from seeking full pseudo masks, another recent work [18] proposed to also use pre-trained edge detector to aid boundary localization. These researches have made commendable contributions.

While pseudo labels may provide richer supervision at unlabeled pixels, the impact from incorrectly generated labels can be reinforced during training [19]. An alternative way is to learn directly towards scribbles using a partial cross-entropy (pCE) loss [20] with aided by energy minimization. For instance, the work in [21] applied conditional random fields (CRF) as a post-processing step to refine model predictions. More efforts have been devoted to integrating the energy term directly into end-to-end training. ScribbleSup [20] and several other works [22–24] had laid the foundations for using CRF and normalized cut as an integrated learning objective. Following this path, some recent works [19,25,26] proposed to unify optimization steps as gradient descent in single-stage training. In particular, the work in [19] showed that using a straightforward CRF regularization could reach the apex in weakly supervised segmentation.

One key reason for the successes of CRF [19] is that it can preserve target boundaries on top of model estimates with shallow vision features, usually image intensity and spatial location. However, handcrafted configurations of gaussian kernels, similarity metrics and kernel sizes could heavily rely on expertise and dataset characteristics. In addition, though CRF, in its formulation, should possess global information, people normally constrain kernel computation within a small neighborhood for efficiency. The current CRF is destined to weigh up computational feasibility and spatial scope of context information. Most deep networks [21,27–29], regardless of their architectures, could effortlessly provide spatial features at different scales and receptive fields. However, it is by no means a trivial task to use these feature layers as part of CRF. Since the networks are trained end-to-end with (partial) segmentation supervision, their internal features were already tailored to the learning target. Therefore, mindlessly feeding feature maps to CRF kernels could intensify incorrect information from the target if the network is not properly rectified. Alternatively, using a separate network to learn representative features for CRF seems plausible, as decoupling from the main backbone creates diversity [6] while avoiding strengthening errors. However, this spells another uncertainty in how to design this auxiliary network, and how much resource overhead is expected.

Motivated by these challenges, we present a straightforward yet effective learning framework. Inspired by recent advances in transformer-based segmentation [30,31], we create a pluggable spatial self-attention module that could be attached on top of any internal feature layers of a fully convolutional network (FCN). Compared to end-to-end transformer architecture, this design infuses global contextual information into the main backbone while keeping the efficiency of convolutions. Descended from this module, we construct a similarity metric based on normalized and symmetrized attention. This attentive similarity, together with a distance mapping mask that encodes spatial locations, further participates in a novel regularization loss that imposes consistency between segmentation prediction and visual feature affinity. Compared to CRF, this loss is efficient since the kernels were pre-computed within the attention module. This attentive similarity objective drives a joint alignment of FCN encoders, attention mapping, and model prediction. Ultimately, the entire end-to-end learning framework can be established with the proposed FCN+Attention architecture and a combination of three objectives: partial segmentation loss, a customized masked CRF loss and attentive similarity loss. The framework is trained in a single stage with scribble annotations, and no pseudo labels or pre-trained feature detectors will be needed.

The contributions of this work are three-fold. (1) We proposed a pluggable spatial self-attention module on top of FCN for scribble-supervised segmentation. The module can work with any backbone architecture and can model interactions of all spatial locations. Stemming from this module, we designed an attentive similarity loss that supports one-stage end-to-end learning. Combining global attention with FCNs is not an entirely new idea [32–34], but to our best knowledge, this is the first work to cultivate a particular design for scribble supervision task, where the novel attentive similarity loss plays a central role in regularization. (2) We customized a masked CRF regularization that drives decent segmentation accuracy and facilitates the capacity of attentive similarity learning. (3) Extensive experiments on public datasets (ACDC and CHAOS) showed that our framework not just out-performs existing state-of-the-art, but also delivers close performance to fully-supervised benchmarks.

2. Method

2.1. Overview

The proposed method is depicted in Fig. 1. The backbone network, exemplified by a U-shaped architecture, is jointly trained with three objectives: partial segmentation loss, masked CRF loss and the affinity similarity loss. Unlike those pseudo label based approaches [6,12–15,18], where pseudo labels contribute to the forward and backward operations, our method ensures that only labeled pixels along scribbles contribute to the partial segmentation loss. In addition, we implement a customized masked CRF regularization, similar to study [19], which effectively suppresses false positives. Simultaneously, we design an attention guided similarity loss to enforce consistency between model predictions and learned visual affinity from encoded features. A selected encoder layer is first processed through a spatial self-attention module, and the learned attention is transformed to create a similarity mask that captures spatial dependencies. This mask is subsequently combined with a distance map encoding spatial locations, ultimately forming the attentive similarity loss. Beyond the learning objectives, the spatial self-attention module contributes to architectural enhancements by introducing global interactions into CNN features.

2.2. Scribble supervision learning

The scribble annotation mask consists of a few partially connected pixels with known class labels $C = \{0, 1, \dots, C\}$, while the remaining pixels remain untagged. Let Ω_L, Ω_U represent the set of labeled and unlabeled pixels, respectively. The label mask is defined as $Y_s[p] = Y[p] + l_U, p \in \Omega_L, p \in \Omega_U$ where $Y[p]$ is the ground truth label at pixel p and $l_U \notin C$ serves as a marker for "unknown". Pseudo label based methods resorted to extend labels from Ω_L to Ω_U. However, incorrect label propagation within Ω_U can lead the model to make inaccurate predictions. Consequently, we directly use Y_s as the sole supervision. One such formulations is the partial cross entropy (pCE) [20] loss:

$$l_{seg}(P, Y_s) = -\frac{1}{|\Omega_L|} \sum_{p \in \Omega_L} \log(P_p^{Y_s[p]}) \tag{1}$$

where P_p^c is the model predicted probability that pixel p belongs to class c.

2.3. Masked conditional random fields

2.3.1. CRF formulation and optimization

Under partial supervision, the network propagates labels to Ω_U based on learned semantic similarities. However, the prediction could be highly inaccurate around object boundaries due to the inherent limitations of scribble labels. The study [35] proposed using label six border points that effectively alleviated boundary ambiguity. However, scribble annotations normally locate around the target skeleton, making it difficult to provide such detailed clues without re-labeling.

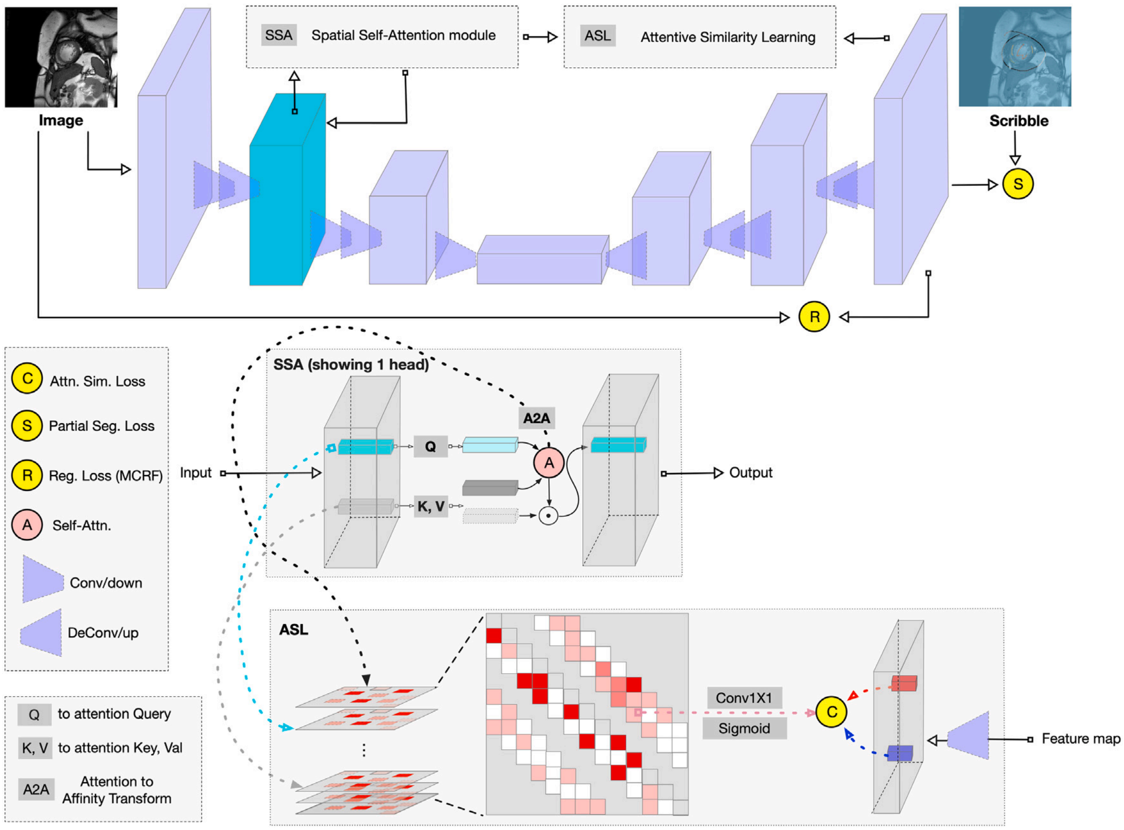

Fig. 1. The figure illustrates the proposed architecture for the end-to-end learning framework that integrates Spatial Self-Attention (SSA) and Attentive Similarity Learning (ASL) with Masked Conditional Random Fields (MCRF). The image undergoes segmentation through a combination of these strategies, with the attention mechanism (Q, K, V) used to optimize feature interactions across spatial regions. The framework includes convolutional, deconvolutional layers, and loss functions (Attn. Sim. Loss, Partial Seg. Loss, Reg. Loss from MCRF) to refine the segmentation and improve performance in weakly supervised learning settings.

Fortunately, CRF have been proven effective for boundary refinement. The formulation of CRF loss is defined as follows:

$$l_{\text{crf}}(P, I) = \frac{1}{|\Omega|} \sum_{p \neq q \in \Omega} \sum_{i \neq j \in C} s(p, q) P_p^i P_q^j \tag{2}$$

where I is the image input and P is the model prediction. $s(p,q)$ is a similarity metric for any pairs of pixels $\{p \neq q\}$.

2.3.2. Integration with segmentation loss

For high-resolution medical images, incorporating all pairs of pixels can be computationally expensive. To address this, it is common practice to impose a limited neighboring window with a fixed radius r. In addition, unwanted pixels can be excluded to avoid introducing noise. The Gated CRF approach [19] used predefined masks to eliminate the influence of image augmentation-induced artifacts. In contrast to study [19], we incorporate extrapolated margins from random rotation augmentation into the loss. This customized mask helps to mitigate false positives near image borders while excluding pixels already annotated by scribbles. The masked CRF loss is then formulated as Eq. (3):

$$l_{\text{mcrf}}(P, I) = \frac{1}{|\Omega_U|} \sum_{p \in \Omega_U} \sum_{q \in \Omega_U : 0 < \|q-p\|_\infty < r} s(p, q) \sum_{i \neq j \in C} P_p^i P_q^j \tag{3}$$

Normally, $s(p,q)$ consists of a weighted combination of multiple Gaussian kernels:

$$s(p, q) = \sum_{k=1}^{K} w^k e^{-\frac{1}{2(\sigma^k)^2} \|f_p - f_q\|_2^2} \tag{4}$$

We found that using just a single kernel that encodes intensity and location similarity $f_p = [I_p, x_p, y_p]^{\mathrm{T}}$ could already improve the segmentation accuracy to a reasonable level.

2.4. Transformer-based spatial self-attention

Inspired by the elegant design proposed by [6], an auxiliary architecture could be employed to provide richer features for the CRF loss. However, a more challenging approach, which is the focus of this work, is to directly leverage the main network without introducing auxiliary overhead. Since the network is trained end-to-end for the segmentation task, the spatial similarity of internal features should already align with those in the model's prediction. If P is not sufficiently enough, re-imposing pairwise (dis-)similarity according to P may exacerbate both correctly and incorrectly predicted pixels. Therefore, instead of directly using internal feature maps, a new design approach is required.

2.4.1. Self-attention mechanism

As mentioned in Section 1, transformers are more effective than CNNs at capturing long-range dependencies due to self-attention, which can be viewed as embedded affinities among pairs of spatial locations. Rather than switching to a transformer-based backbone, we integrate self-attention into a pluggable module that can be attached to any feature layer of arbitrary CNN architectures. Let $f \in \mathbb{R}^{hw \times d}$ represent the flattened feature map, where h and w denote the spatial resolution and d the feature dimension. The ith attention head is constructed by three separate linear transforms, producing query Q_i, key K_i and value V_i. The attention is then computed as $A_i = Q_i K_i^{\mathrm{T}} / \sqrt{d_k}$ and the attended feature is given by $a_i = \mathrm{softmax}(A_i) V_i$. The concatenated multi-head features $a = \mathrm{concat}(a_i)$ are then passed through layer normalization and MLP with skip connections: $a = a + \mathrm{MLP}_{d_v \to d_v}(\mathrm{LayerNorm}(a))$. If there are multiple identical layers of these operations in a sequence, the output feature of the current block becomes the input of the next.

2.4.2. Enhancing spatial dependencies

In our context, the attended feature will eventually need to be compressed back to the original dimension using $a = \mathrm{MLP}_{ndv\to d}(a)$ to facilitate subsequent convolutions. Such a seemingly straightforward module brings two benefits: first, it encapsulates global long-term dependencies within the CNN backbone, and thus increased the scope of receptive field to the entire spatial domain. Second, the learned attention A_i provides valuable information about pairwise similarity among different spatial locations. By leveraging A_i in a regularization loss, such as in Eq. (3), we can obtain $s(p,q)$ without the computational burden of $O(h^2w^2)$ complexity.

2.5. Attentive similarity learning mechanism

This section will describe how to transform the attention into a regularization loss and why our design could work in a way free from the earlier mentioned paradox.

2.5.1. Similarity metric design

Similar to attended features, we concatenate A_i from all $n \geq 1$ heads and all $L \geq 1$ layers: $A = \mathrm{concat}(A_i) \in \mathbb{R}^{hw\times hw\times nL}$. Since Q and K are different transforms, A is normally asymmetric. Next, we apply a softmax normalization to the rows of A for the reasons to be explained later: $A = \mathrm{softmax}(A)$. We then derive a symmetric version of $\bar{A} = A + A^T \in \mathbb{R}^{hw\times hw\times nL}$ where the transpose is applied ot the first two dimensions. Finally, a 1×1 convolution is used to reduce the channel dimension and transform $\bar{A}$ into pairwise similarity metric: $S = \mathrm{sigmoid}(\mathrm{Conv}^{1\times 1}_{nL\to 1}(\bar{A})) \in \mathbb{R}^{hw\times hw}$.

It may seem convenient to directly use $s(p,q) = S[p,q]$ in Eq. (3) to replace the hand-crafted kernel. However, this approach would degrade model performance because S does not capture effective positional information. Although convolution operation with zero-padding can implicitly leak partial location signals [30], it is not sufficient in our context, which requires accurate distance mapping. Without the inclusion of x, y coordinates, there is a risk of introducing spurious false positives at locations with large similarity values in S, but not relevant context.

2.5.2. Loss function implementation

We encode location information explicitly with a distance decay mapping: $\mathcal{M} \in \mathbb{R}^{hw\times hw}$ with $\mathcal{M}[p,q] = \exp\{-\|p_{x,y} - q_{x,y}\|_2^2/2\sigma^2\}$. In addition, the network layer attached to the attention module normally has a lower resolution than predicted mask P, in this case, we downsample P to $h \times w$: $\tilde{P} = \mathrm{Interpolate}(P, h\times w)$. To this end, we have the regularization loss from the attention module in Eq. (5):

$$l_{\mathrm{atn}}(P,S) = \frac{1}{|\Omega_U|}\sum_{p\in\Omega_U}\sum_{q\in\Omega_U:0<\|q-p\|_\infty<r}\mathcal{M}[p,q]S[p,q]\sum_{i\neq j\in C}\tilde{P}^i_p\tilde{P}^j_q \tag{5}$$

Although this loss adopts a similar form, it fundamentally differs from the Masked CRF loss described in Eq. (3). It involves the joint optimization of P and S, where both the main network and the attention branch receive gradient updates during back-propagation. An improved P will lead to better production of S, which in turn refines the representations in the earlier features and attention. Moreover, the learnable parameters in attention block foster information diversity, which prevents the downward spiral of reinforcing incorrect predictions. By suppressing similarities between pairs with distant P, the model automatically encourages higher similarity among pairs with closer P. In contrast, a naive optimization approach would set all $S[p,q]$ to zero. In summary, this attentive similarity loss addresses the challenge of designing CRF-like regularization within a single network. We will demonstrate its effectiveness in experiments.

Ultimately, the network will be trained by optimizing the following objective [36]:

$$l = l_{\mathrm{seg}}(P, Y^s) + \lambda_{\mathrm{mcrf}} l_{\mathrm{mcrf}}(P, I) + \lambda_{\mathrm{atn}} l_{\mathrm{atn}}(P,S) \tag{6}$$

3. Experiment and results

3.1. Datasets

We evaluate the proposed method and existing state-of-the-arts methods on two datasets, ACDC [6,36] and CHAOS [37,38]. The ACDC dataset includes 200 3D cine-MRI images obtained from 100 patients (two images from each patient). Ground truth annotations are provided for end-diastolic (ED) and end-systolic (ES) cardiac phases for three foreground targets: right ventricle (RV), myocardium (Myo), and left ventricle (LV). Scribble annotations were directly adopted from a recent release in [38]. The CHAOS dataset has 20 3D abdominal MR images of 20 subjects (one image per subject), the full segmentation masks are provided for liver, kidneys, and spleen. We choose the T1 in-phase in this paper. Since no scribbles were provided, we simulated scribble annotations for foreground and background classes similar to the steps in [38].

To acquire the scribble annotation in CHAOS T1 Phase, we followed the methods outlined in the research article by Valvano et al. [38]. The scribble annotations for each organ were obtained through standard skeletonization and interactive erosion techniques, whereby the border pixels were continuously eroded until the point of loss of connectivity. Additionally, to create a boundary that closely approximates the periphery of aimed object and to reduce the cost of model learning error correction, the skeleton of the outside boundary was constrained. Specifically, we generated the largest convex hull that could encapsulate all organs and then expanded the convex hull by 5 pixels to serve as an additional boundary for the overlapping background area. Pixels falling outside of these defined boundaries were left unclassified as belonging to either organ or background. The generated scribble data will be released upon publication. For both dataset, we split it into training and evaluation sets by patient. For ACDC, 20 patients (40 images) were allocated for evaluation and the rest 80 patients (160 images) for training. For CHAOS, 7 subjects/images were used for evaluation and the rest 14 subjects/images for training. All methods will be trained and tested on the identical set-up. For reproducibility, the patient ids corresponding to this partition will be given upon code release.

3.2. Implementation

We choose 2D U-Net [27] as opposed to 3D FCNs as the backbone network. Since that on both datasets the thickness between slices is ~6 times larger than pixel spacing, also that scribbles were delineated on 2D slices instead of 3D volumes [6,38]. The proposed self-attention module was grafted onto the third convolutional layer of encoder. All experiments were run with PyTorch 1.7.1 environment on 8 RTX A6000 GPUs. During training, we standardize image intensity to the range of $[0,1]$ for each 3D volume. Then, each slice goes through augmentation steps such as random rotation/flipping following a resizing to 256×256. This process is similar to [6], but we took more stringent augmentations on CHAOS since flipping and 90° rotation could confuse left/right kidneys. The network is trained end-to-end with stochastic gradient descent and polynomial learning rate decay schedule. We set both λ_{atn} and λ_{mcrf} to 0.1. During testing, 2D predictions were stacked to produce 3D masks. No post-refinement steps were adopted. The 3D Dice Coefficient and 95% (DSC) Hausdorff Distance (HD95) are used for evaluation. Note that HD95(mm) should be computed with pixel spacing and thickness considered.

3.3. Results

3.3.1. Comparing with other methods

We compare our method with 8 existing methods [6,13,14,19,25,26,39,40] for scribble supervised segmentation and the supposed

Table 1
Eval. on ACDC. Displayed `mean(std)` of DSC and HD95(mm) across all test samples, for three different foreground classes (RV, Myo, LV) respectively as well as avg. across all classes. Top 2 best metrics (DSC↑, HD95↓) including ties will be shown in **bold**, top metric without ties will be shown in ***bold-italic***.

Method	RV		Myo		LV		Avg.	
	DSC	HD95	DSC	HD95	DSC	HD95	DSC	HD95
Fully	0.89(0.079)	10.47(20.17)	0.88 (0.028)	2.21(1.91)	0.96 (0.009)	2.21(2.07)	0.91 (0.031)	4.96(6.97)
pCE [20]	0.58(0.16)	186.25(28.7)	0.48(0.12)	166.67(17.67)	0.79(0.10)	167.41(18.24)	0.62(0.10)	173.44(18.56)
RW [13]	0.82(0.11)	12.7(24.72)	0.7(0.049)	8.01(2.43)	0.89(0.036)	10.4(14.04)	0.8 (0.043)	10.37(12.97)
USTM [39]	0.75(0.13)	124.93(64.88)	0.69(0.079)	141.66(24.83)	0.82(0.068)	153.36(28.2)	0.75(0.069)	139.98(30.25)
S2L [14]	0.84(0.095)	17.61(31.27)	0.8(0.056)	46.52(53.71)	0.92(0.039)	53.31(58.32)	0.86(0.041)	39.14(37.07)
MS [25]	0.86(0.075)	11.31(18.85)	**0.83(0.043)**	14.14(34.46)	**0.94(0.027)**	17.42(38.39)	**0.88(0.033)**	14.29(25.69)
EM [22]	0.83(0.83)	24.07(42.55)	0.81(0.051)	44.18(51.55)	0.92(0.035)	36.19(48.14)	0.85(0.043)	34.81(35.71)
GCRF [19]	0.87(0.083)	11.7(21.51)	**0.83(0.036)**	15.76(36.28)	**0.94(0.018)**	13.64(35.06)	**0.88(0.034)**	13.70(24.77)
Contour [26]	0.61(0.12)	192.85(27.55)	0.56 (0.084)	162.88(16.81)	0.79(0.066)	172.08(20.96)	0.65(0.071)	175.94 (18.50)
DMPLS [6]	0.86 (0.085)	15.98(30.52)	**0.83(0.041)**	20.72(40.81)	0.93 (0.023)	24.51(43.77)	0.87(0.034)	20.40(30.45)
ScribbleVC [16]	0.87(0.073)	8.76(15.6)	0.80(0.046)	11.47(33.73)	0.93(0.019)	11.31(33.84)	0.87(0.030)	10.51(22.93)
ScribFormer [17]	***0.88(0.084)***	***7.25(15.4)***	**0.83(0.048)**	13.34(36.36)	0.93(0.025)	16.1(38.56)	**0.88(0.036)**	12.23(26.513)
Ours w/o ASL	0.87(0.097)	13.79(29.16)	**0.83(0.042)**	***3.45(1.99)***	**0.94(0.027)**	***4.16(6.38)***	**0.88(0.040)**	***7.13(10.02)***
Ours	**0.87(0.079)**	10.72(19.34)	**0.83(0.038)**	***3.29(1.84)***	**0.94(0.016)**	***5.13(8.86)***	**0.88(0.032)**	***6.38(8.53)***

Table 2
Eval. on CHAOS. Displayed `mean(std)` of DSC and HD95(mm) across all test samples, for four different foreground classes (LIV, RK, LK, SPL) respectively as well as avg. across all classes. Top 2 best metrics (DSC↑, HD95↓) including ties will be shown in **bold**, top metric without ties will be shown in ***bold-italic***.

Method	LIV		RK		LK		SPL		Avg.	
	DSC	HD95	DSC	HD95	DSC	HD95	DSC	HD95	DSC	HD95
Fully	0.92(0.016)	10.0(3.88)	0.9(0.030)	20.72(16.89)	0.9(0.061)	13.2(14.12)	0.83(0.090)	12.2(8.35)	0.89(0.041)	14.03(6.20)
pCE [20]	0.85(0.025)	12.24(2.13)	0.71(0.095)	51.28(39.81)	0.72(0.110)	46.48(31.35)	0.71(0.1)	39.19(14.22)	0.75(0.076)	37.3(11.90)
RW [13]	0.762(0.028)	22.74(15.54)	0.634(0.101)	44.57(19.93)	0.676(0.152)	18.05(9.95)	0.725(0.120)	30.94(19.47)	0.699(0.090)	29.07(6.41)
USTM [39]	0.81(0.024)	13.78(0.69)	0.7(0.088)	30.29(12.45)	0.72(0.12)	45.76(34.27)	0.69(0.19)	33.32(20.28)	0.73(0.1)	30.79(13.49)
S2L [14]	0.82(0.024)	34.73(16.65)	0.7(0.065)	72.74(56.45)	0.67(0.14)	92.38(67.07)	0.64(0.21)	84.91(28.33)	0.71(0.09)	71.19(15.45)
MS [25]	0.64(0.035)	25.2(1.42)	0.65(0.068)	54.13(18.50)	0.63(0.1)	34.96(31.64)	0.62(0.21)	25.59(20.13)	0.64(0.089)	34.97(14.22)
EM [22]	0.84(0.022)	11.51(0.88)	0.69(0.084)	28.35(14.39)	0.71(0.12)	39.31(43.56)	0.71(0.14)	28.38(11.69)	0.74(0.083)	26.89(11.28)
GCRF [19]	**0.92(0.022)**	**8.97(2.61)**	**0.87(0.043)**	25.02(15.63)	**0.89(0.053)**	10.79(9.19)	**0.85(0.062)**	**11.13(7.37)**	**0.88(0.037)**	**13.98(4.44)**
Contour [26]	0.84(0.022)	13.29(2.12)	0.66(0.1)	23.84(13.83)	0.71(0.15)	14.85(8.13)	0.72(0.12)	28.72(21.97)	0.73(0.096)	20.18(7.01)
DMPLS [6]	0.89(0.020)	11.19(4.09)	0.85(0.028)	32.33(25.68)	0.84(0.082)	15.13(10.56)	0.79(0.12)	12.77(7.65)	0.84(0.054)	17.85(6.99)
Ours w/o ASL	**0.92(0.018)**	14.46(8.65)	0.86(0.030)	***19.79(14.66)***	**0.89(0.038)**	***8.55(6.06)***	0.82(0.1)	13.18(7.91)	**0.87(0.041)**	**13.99(5.51)**
Ours	***0.93(0.017)***	***8.19(1.56)***	**0.87(0.030)**	***18.55(17.57)***	**0.89(0.052)**	***7.79(5.23)***	**0.84(0.089)**	***10.21(8.18)***	**0.88(0.040)**	***11.18(5.00)***

lower/upper benchmarks. In particular, training the network with partial cross entropy (pCE) [20] loss alone can be treated as a baseline benchmark. At the other end of the spectrum, training with fully annotated masks (Fully) should provide upper bound performance since no regularization/pseudo label could produce a stronger supervision than pixel-level ground truth.

Among the 8 methods, RW (random walker) [13], USTM [39], S2L (Scribble2Label) [14] and the most recent DMPLS [6] leveraged pseudo labels in one way or another. In particular, both USTM [39] and DMPLS [6] adopted double network/branches to promote robustness. Besides, the rest of four methods, MS (Mumford-Shah loss) [25], EM (Entropy Minimization) [40], GCRF (GatedCRFLoss) [19] and Contour (contour loss) [26] explored regularization terms with pCE training. Another recent work, Scribble2D5 [18] also showed exceptional performances, however, it is not included in our comparison since they rely on external dataset to pre-train an edge detector, which may induce uncertainty in both fairness and interpretability of the experiment.

Tables 1 and 2 show the results on ACDC and CHAOS respectively. For both datasets, our framework (last row in the tables) delivers the best DSC and HD95. Some methods such as DMPLS [6], ScribFormer [17] and GCRF [19] give the same level of DSC on particular organs, but our HD95 is significantly lower than theirs. The ScribbleVC [16], ScribFormer [17], and GCRF [19] methods demonstrate effective control over distance-based metric HD95, indicating that they are capable of extracting structural information from the images effectively.

For CHAOS, our method can even produce lower HD95 and the same level of DSC (on LIV, LK, SPL and average) compared to the fully supervision benchmark. This indicates that with masked CRF and attentive similarity learning, our method is better at preserving target boundaries and reducing false positives. This conclusion can be further verified in Section 3.4. For medical images, precisely predicting object boundaries is critical in extracting subtle 3D structures with clinical significance. We also observed that in both tables, Gated CRF [19] and our method yield equal DSC. This may be caused by two reasons that also reveal the inherent mechanisms: (1) limited by sparse annotation, there is an upper limit of DSC below fully supervised benchmark; the tables (especially Table 2) show that both methods approach this upper limit, and therefore the similar DSC. (2) the role of masked CRF (Section 2.3) which resembles Gated CRF [19] should be to push the prediction accuracy to a reasonable level, upon which attentive similarity learning (Section 2.5) takes over to optimize the boundaries. The ASL module is fundamentally different from masked CRF, and cannot be used to replace CRF. This verified our statements in Section 2.5.

3.3.2. Ablation study

It might seem reasonable to investigate individually the importance of three essential modules, SSA (Section 2.4), MCRF (Section 2.3) and ASL 2.5 of our framework. The three modules need to work together to guarantee the top performance. MCRF helped to boost the accuracy in target localization and shape (measured by DSC), then SSA and ASL could then show their capacities in reducing false-positives and preserving boundaries. In addition, there is no way to keep ASL without SSA since ASL was directly derived from SSA.

Therefore, we only examine how the performance could change if we remove ASL. From both tables (second row from bottom), removing ASL would make HD95 worse, for all organs and avg. with one exception for LV (shown on Table 1). This further verifies our statement about ASL's role on boundary preservation.

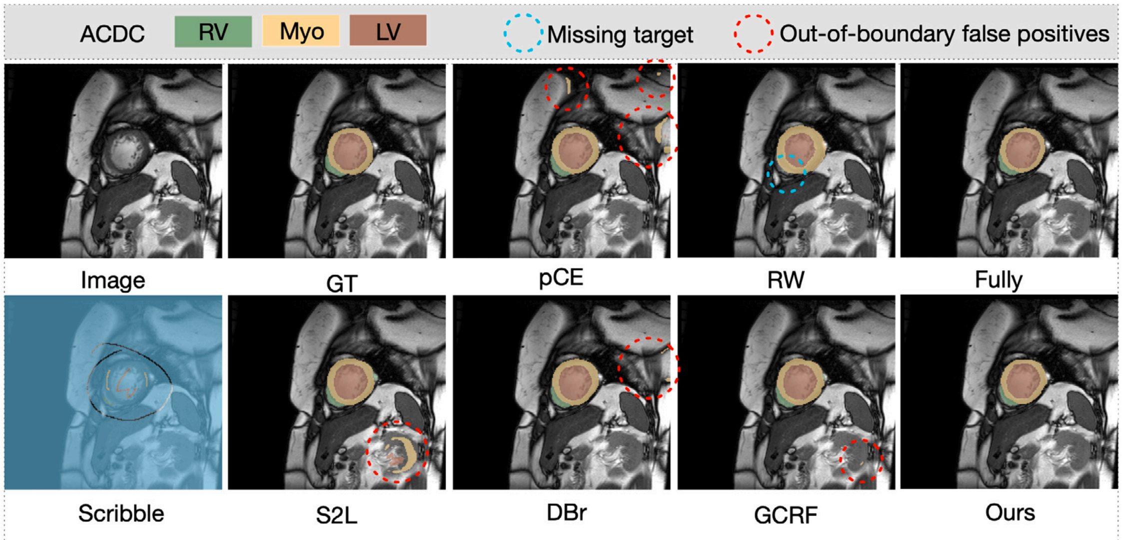

Fig. 2. Segmentation predictions on ACDC slice, GT is ground truth mask. We identify with circles significant false-positives and missing areas (false-negatives).

Table 3
Ablation experiments on SSA for comparing placing SSA at different Encoder layers and evaluating DSC and HD95(mm) performance on ACDC dataset.

Models	RV	MYO	LV	Avg. DSC	Avg. HD95
SSA at E2	**0.87**	**0.83**	**0.94**	**0.88**	**6.38**
SSA at E3	0.86	0.81	0.94	0.87	11.04
SSA at E4	0.87	0.82	0.93	0.87	9.23

Table 4
Ablation experiments for different modules, including the masked CRF, spatial self-attention, and the attentive similarity learning module. The performance was evaluated on the ACDC dataset using DSC scores and HD95 (mm) metrics.

Models	MCRF	SSA	ASL	RV	MYO	LV	Avg. DSC	Avg. HD95
#1	✓	✓	×	0.87	0.83	0.94	0.88	7.13
#2	✓	×	×	0.86	0.82	0.94	0.88	11.87
#3	×	✓	✓	0.79	0.75	0.90	0.81	63.30
#4	✓	✓	✓	**0.87**	**0.83**	**0.94**	**0.88**	**6.38**

Moreover, even without ASL, our DSC and HD95 still stay on the best level for most part of ACDC and CHAOS, with three exceptions including ACDC-RV, CHAOS-LIV and CHAOS-SPL. This further indicates that both SSA and ASL contributed to reducing boundary inconsistencies.

The ablation results of SSA module placing in different layers are presented in Table 3. Due to the constraints of GPU memory, it was not feasible to compare SSA at E1 under identical conditions. To accommodate this module, the batch size and the number of channels would have required reduction, and thus this comparison was not conducted. The results in Table 3 indicate that the SSA layer placed in the second encoder layer allows the model to obtain higher DSC and lower HD95 performances.

As the proposed framework integrates MCRF, SSA, and ASL modules, we conducted ablation experiments to compare their contributions to the segmentation performance in the proposed framework. As shown in Table 4, MCRF provides the most substantial improvement in DSC under scribble supervision, while SSA and ASL notably enhance the model's HD95 metrics.

For comparison, the paired Samples t-test was conducted on the segmentation metric results of different segmentation methods. The results are summarized in Table 5. The results on ACDC dataset indicate that, except for comparisons with the MS, ScribFormer and DMPLS methods, the p-values of DSC for proposed segmentation method are less than 0.05 when compared with other methods. And all the p-values of HD95 indicated that the differences were significant. The DSC results on CHAOS dataset show that, except for the GCRF and Fully Segmentation methods, the p-values for our proposed method are also less than 0.05 when compared to the other segmentation methods. The performance improvement of our model on the CHAOS dataset is not significant compared to other SOTA methods, probably due to the limited number of samples and limited boosts. However, these suggest that our method performs comparably well to these excellent approaches and has the potential to approximate the segmentation performance of fully supervised methods.

3.4. Qualitative analysis

We visualize segmentation results along with scribbles/ground truth labels from representative 2D slices in Figs. 2 and 3. In Fig. 2, our fully supervised model provides the most similar predictions to the ground truth. In addition, false positives can be effectively mitigated by our approach compared to other methods. This in turn demonstrates the conclusion observed in Tables 1 and 2. In Fig. 3, we see that aside from suppressing false positives, our method also preserves target boundaries better while all other methods give cupped boundaries for some targets. Compared to fully supervised learning, our method missed out on a small part of LK, but sufficiently predicted the tipping part of LIV missed out by Fully. Finally, in order to explore the effect of different hyperparameter settings on the model segmentation performance, we adjusted parameters λ_{atn} and λ_{mcrf}, and obtained the performance of the model using evaluation metrics DSC and HD95, which are shown in Fig. 4. From comparison results, for both datasets, metrics are more robust against changes of λ_{atn}.

4. Discussion and conclusion

In this study, a novel framework was proposed for improving the segmentation performance, especially the distance-based metric, in the scribble supervised medical image segmentation. We introduced an end-to-end learning framework incorporating spatial self-attention (SSA), masked conditional random field (MCRF) and attentive similarity learning (ASL). The SSA module can be integrated into any internal layer of fully convolutional network, enhancing representation power by introducing global interactions. Attentive similarity learning branches from SSA to enforce consistency between visual feature affinity and predictions. Additionally, MCRF contributes to improve

Fig. 3. Segmentation predictions on CHAOS slice, GT is ground truth mask. We identify with circles significant false-positives and missing areas (false-negatives).

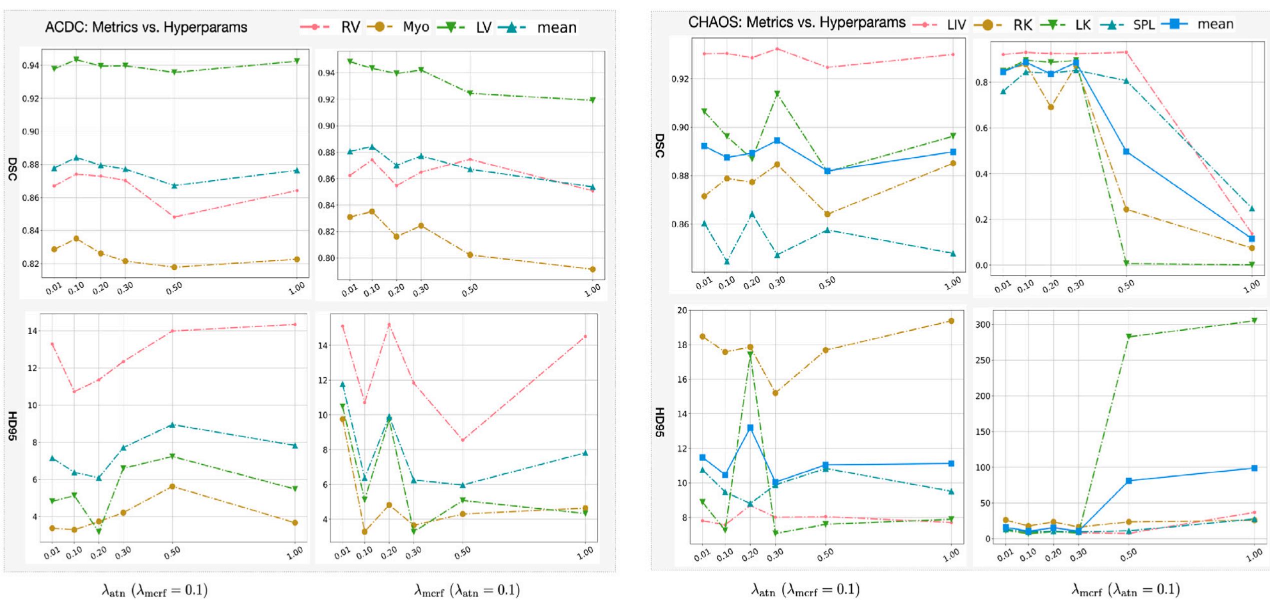

Fig. 4. Sensitivity analysis of hyperparameters on the proposed method: λ_{atn}, λ_{mcrf}; when varying one of them, the other was kept as 0.1.

Table 5
The p-values from the pair Student's t-test comparing segmentation performance of proposed and others' segmentation method.

	ACDC		CHAOS	
	DICE	HD95	DICE	HD95
pCE	1.957×10^{-19}	7.238×10^{-19}	4.175×10^{-4}	8.023×10^{-4}
RW	3.120×10^{-19}	7.707×10^{-19}	2.305×10^{-4}	1.509×10^{-2}
USTM	3.570×10^{-16}	1.239×10^{-9}	2.212×10^{-3}	5.053×10^{-3}
S2L	1.465×10^{-5}	3.280×10^{-13}	7.827×10^{-4}	3.511×10^{-6}
MS	8.204×10^{-1}	9.497×10^{-20}	4.933×10^{-5}	3.851×10^{-4}
EM	3.371×10^{-5}	1.599×10^{-13}	7.092×10^{-4}	1.295×10^{-3}
GCRF	2.830×10^{-2}	2.607×10^{-20}	7.614×10^{-2}	5.985×10^{-1}
Contour	3.603×10^{-22}	1.213×10^{-19}	1.882×10^{-3}	3.217×10^{-1}
DMPLS	8.602×10^{-1}	8.376×10^{-17}	1.075×10^{-2}	6.637×10^{-1}
ScribbleVC	6.355×10^{-3}	2.477×10^{-21}	–	–
ScribFormer	9.123×10^{-2}	1.079×10^{-19}	–	–
Fully	1.055×10^{-16}	3.219×10^{-20}	6.291×10^{-1}	6.507×10^{-1}

segmentation accuracy, thereby enhancing the effectiveness of SSA and ASL. Experiments conducted on two widely used public datasets, ACDC and CHAOS, demonstrated that the proposed method outperformed nine popular approaches in terms of overall performance, particularly in significantly reducing HD95 values. Based on the results in this study, the following paragraph will discuss the ASL and GCRF strategies.

First, ASL strategy to segmentation in this study is relatively small in DSC metric when used in isolation. However, this module primarily contributes to the overall segmentation model when combined with other modules. Second, the main design of the ASL module is to transform attention into a regularization loss function, which assists in optimizing the segmentation model. On the ACDC and CHAOS datasets, the model's performance on HD95 metric showed greater improvement when ASL was added, with improvements from 7.13 mm to 6.38 mm (−10.5%) and from 13.99 mm to 11.18 mm (−20.0%), respectively. The ASL module indeed contributes to segmentation, particularly in distance metrics.

GCRF is a well-designed algorithm that uses distance functions to constrain segmentation results and optimize the model. The reason why our proposed method shows limited improvement over GCRF is that, in the current experiments, the performance of the methods evaluated, in terms of DSC and HD95 metrics, is already very close to that of fully supervised segmentation results. Fully supervised segmentation results can be considered as the upper bound of weakly supervised segmentation performance, and as such, the current methods show performance that is close to the fully supervised approach, leaving little room for further improvement beyond this benchmark. Consequently, the segmentation performance of our proposed model is similar to that of the GCRF model.

Compared to previous studies, we have developed a weakly supervised segmentation model based on a transformer-based framework, primarily leveraging loss functions rather than pseudo label to improve segmentation performance. In contrast to fully supervised segmentation models, the weakly supervised annotation approach reduces the labor costs associated with data labeling. This approach could be more conducive to training data-driven models and may serve as a useful reference for future research. Our current results suggest that the distance-base loss function constraint plays a crucial role, and we hope that future work will focus on further advancing the design of loss functions to provide greater interpretability for the model.

However, the current study has the following limitations. (1) The training and validation data are restricted to CT and MRI images, and do not include more complex and colorful modalities such as laparoscopic and dermoscopic images. Expanding the dataset to incorporate these types of images could further validate the generalizability and robustness of the proposed method. (2) The improvement in Dice Score is not substantial, which is partly due to the limited amount of annotated data used in the current study. Future work will involve comparing the performance of the proposed method with a larger set of annotated samples to assess its effectiveness more comprehensively. (3) The current model is constrained to 2D image processing, and the GPU memory limitations restrict the model's ability to handle complete 3D image data. In future iterations, we aim to reduce the model's parameter size and explore 3D model annotations to mitigate the limitations of processing images layer by layer in 2D.

In conclusion, our scribble-based supervised AI method achieves segmentation results close to fully supervised annotations, providing a promising tool for enhancing relational segmentation in medical images processing. Future research can further refine and expand this method to ensure more reliable and accurate results in medical images studies. Additionally, applying these methods to broader range of weakly supervised scenarios could improve segmentation and annotation efficiency.

CRediT authorship contribution statement

Mu Tian: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Qinzhu Yang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Yi Gao:** Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] Nima Tajbakhsh, Laura Jeyaseelan, Qian Li, Jeffrey N. Chiang, Zhihao Wu, Xiaowei Ding, Embracing imperfect datasets: A review of deep learning solutions for medical image segmentation, Med. Image Anal. 63 (2020) 101693.

[2] Yuyin Zhou, Zhe Li, Song Bai, Chong Wang, Xinlei Chen, Mei Han, Elliot Fishman, Alan L. Yuille, Prior-aware neural network for partially-supervised multi-organ segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 10672–10681.

[3] Anna Khoreva, Rodrigo Benenson, Jan Hosang, Matthias Hein, Bernt Schiele, Simple does it: Weakly supervised instance and semantic segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 876–885.

[4] Yigit B. Can, Krishna Chaitanya, Basil Mustafa, Lisa M. Koch, Ender Konukoglu, Christian F. Baumgartner, Learning to segment medical images with scribble-supervision alone, in: Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 20 2018, Proceedings 4, Springer, 2018, pp. 236–244.

[5] Nasim Souly, Concetto Spampinato, Mubarak Shah, Semi supervised semantic segmentation using generative adversarial network, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 5688–5696.

[6] Xiangde Luo, Minhao Hu, Wenjun Liao, Shuwei Zhai, Tao Song, Guotai Wang, Shaoting Zhang, Scribble-supervised medical image segmentation via dual-branch network and dynamically mixed pseudo labels supervision, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2022: 25th International Conference, Singapore, September (2022) 18–22, Proceedings, Part I, Springer, 2022, pp. 528–538.

[7] Ke Zhang, Xiahai Zhuang, A new pu learning framework regularized by global consistency for scribble supervised cardiac segmentation, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2022: 25th International Conference, Singapore, September (2022) 18–22, Proceedings, Part VIII, Springer, 2022, pp. 162–172.

[8] Zefan Yang, Di Lin, Dong Ni, Yi Wang, Non-iterative scribble-supervised learning with pacing pseudo-masks for medical image segmentation, 2022, arXiv preprint arXiv:2210.10956.

[9] Jifeng Dai, Kaiming He, Jian Sun, Boxsup: Exploiting bounding boxes to supervise convolutional networks for semantic segmentation, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1635–1643.

[10] Hongjun Chen, Jinbao Wang, Hong Cai Chen, Xiantong Zhen, Feng Zheng, Rongrong Ji, Ling Shao, Seminar learning for click-level weakly supervised semantic segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 6920–6929.

[11] Zhang Chen, Zhiqiang Tian, Jihua Zhu, Ce Li, Shaoyi Du, C-cam: Causal cam for weakly supervised semantic segmentation on medical image, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11676–11685.

[12] Dong-Hyun Lee, et al., Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks, in: Workshop on Challenges in Representation Learning, ICML, Vol. 3, 2013, p. 896.

[13] Leo Grady, Random walks for image segmentation, IEEE Trans. Pattern Anal. Mach. Intell. 28 (11) (2006) 1768–1783.

[14] Hyeonsoo Lee, Won-Ki Jeong, Scribble2label: Scribble-supervised cell segmentation via self-generating pseudo-labels with consistency, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October (2020) 4–8, Proceedings, Part I 23, Springer, 2020, pp. 14–23.

[15] Ke Zhang, Xiahai Zhuang, Cyclemix: A holistic strategy for medical image segmentation from scribble supervision, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11656–11665.

[16] Zihan Li, Yuan Zheng, Xiangde Luo, Dandan Shan, Qingqi Hong, Scribblevc: Scribble-supervised medical image segmentation with vision-class embedding, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 3384–3393.

[17] Zihan Li, Yuan Zheng, Dandan Shan, Shuzhou Yang, Qingde Li, Beizhan Wang, Yuanting Zhang, Qingqi Hong, Dinggang Shen, Scribformer: Transformer makes cnn work better for scribble-based medical image segmentation, IEEE Trans. Med. Imaging (2024).

[18] Qiuhui Chen, Yi Hong, Scribble2d5: Weakly-supervised volumetric image segmentation via scribble annotations, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2022: 25th International Conference, Singapore, September (2022) 18–22, Proceedings, Part VIII, Springer, 2022, pp. 234–243.

[19] Anton Obukhov, Stamatios Georgoulis, Dengxin Dai, Luc Van Gool, Gated crf loss for weakly supervised semantic image segmentation, 2019, arXiv preprint arXiv:1906.04651.

[20] Di Lin, Jifeng Dai, Jiaya Jia, Kaiming He, Jian Sun, Scribblesup: Scribble-supervised convolutional networks for semantic segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 3159–3167.

[21] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, Alan L. Yuille, Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, IEEE Trans. Pattern Anal. Mach. Intell. 40 (4) (2017) 834–848.

[22] Meng Tang, Abdelaziz Djelouah, Federico Perazzi, Yuri Boykov, Christopher Schroers, Normalized cut loss for weakly-supervised cnn segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 1818–1827.

[23] Meng Tang, Federico Perazzi, Abdelaziz Djelouah, Ismail Ben Ayed, Christopher Schroers, Yuri Boykov, On regularized losses for weakly-supervised cnn segmentation, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 507–522.

[24] Olga Veksler, Regularized loss for weakly supervised single class semantic segmentation, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August (2020) 23–28, Proceedings, Part XXIX 16, Springer, 2020, pp. 348–365.

[25] Boah Kim, Jong Chul Ye, Mumford–shah loss functional for image segmentation with deep learning, IEEE Trans. Image Process. 29 (2019) 1856–1866.

[26] Rosana E.L. Jurdi, Caroline Petitjean, Paul Honeine, Veronika Cheplygina, Fahed Abdallah, A surprisingly effective perimeter-based loss for medical image segmentation, in: Medical Imaging with Deep Learning, PMLR, 2021, pp. 158–167.

[27] Olaf Ronneberger, Philipp Fischer, Thomas Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October (2015) 5-9, Proceedings, Part III 18, Springer, 2015, pp. 234–241.

[28] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, Jianming Liang, Unet++: Redesigning skip connections to exploit multiscale features in image segmentation, IEEE Trans. Med. Imaging 39 (6) (2019) 1856–1867.

[29] Fabian Isensee, Paul F. Jaeger, Simon A.A. Kohl, Jens Petersen, Klaus H. Maier-Hein, Nnu-net: a self-configuring method for deep learning-based biomedical image segmentation, Nature Methods 18 (2) (2021) 203–211.

[30] Robin Strudel, Ricardo Garcia, Ivan Laptev, Cordelia Schmid, Segmenter: Transformer for semantic segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 7262–7272.

[31] Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger R. Roth, Daguang Xu, Unetr: Transformers for 3d medical image segmentation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 574–584.

[32] X. Wang, Ross B. Girshick, Abhinav Kumar Gupta, Kaiming He, Non-local neural networks, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2017, pp. 7794–7803.

[33] Xiangtai Li, Li Zhang, Ansheng You, Maoke Yang, Kuiyuan Yang, Yunhai Tong, Global aggregation then local distribution in fully convolutional networks, in: British Machine Vision Conference, 2019.

[34] Zilong Huang, Xinggang Wang, Lichao Huang, Chang Huang, Yunchao Wei, Humphrey Shi, Wenyu Liu, Ccnet: Criss-cross attention for semantic segmentation, in: 2019 IEEE/CVF International Conference on Computer Vision, ICCV, 2018, pp. 603–612.

[35] Shuwei Zhai, Guotai Wang, Xiangde Luo, Qian Yue, Kang Li, Shaoting Zhang, Pa-seg: Learning from point annotations for 3d medical image segmentation using contextual regularization and cross knowledge distillation, 2022, ArXiv, abs/2208.05669.

[36] Olivier Bernard, Alain Lalande, Clement Zotti, Frederick Cervenansky, Xin Yang, Pheng-Ann Heng, Irem Cetin, Karim Lekadir, Oscar Camara, Miguel Angel Gonzalez Ballester, et al., Deep learning techniques for automatic mri cardiac multi-structures segmentation and diagnosis: is the problem solved? IEEE Trans. Med. Imaging 37 (11) (2018) 2514–2525.

[37] A. Emre Kavur, N. Sinem Gezer, Mustafa Barış, Sinem Aslan, Pierre-Henri Conze, Vladimir Groza, Duc Duy Pham, Soumick Chatterjee, Philipp Ernst, Savaş Özkan, et al., Chaos challenge-combined (ct-mr) healthy abdominal organ segmentation, Med. Image Anal. 69 (2021) 101950.

[38] Gabriele Valvano, Andrea Leo, Sotirios A. Tsaftaris, Learning to segment from scribbles using multi-scale adversarial attention gates, IEEE Trans. Med. Imaging 40 (8) (2021) 1990–2001.

[39] Xiaoming Liu, Quan Yuan, Yaozong Gao, Kelei He, Shuo Wang, Xiao Tang, Jinshan Tang, Dinggang Shen, Weakly supervised segmentation of covid19 infection with scribble annotation on ct images, Pattern Recognit. 122 (2022) 108341.

[40] Yves Grandvalet, Yoshua Bengio, Semi-supervised learning by entropy minimization, in: ConfÉrence Francophone sur L'Apprentissage Automatique, 2004.